

ly speaking, CD quality comes to 2 Mbps or so, because of additional bits required for error correction and metadata), but your MP3s are typically 128 kbps! Even 256 kbps comes nowhere close to 1378 kbps.

Why, then, do 128 kbps MP3s sound OK? It's because of certain limitations—not everyone can listen very well beyond 10 kHz, and not everyone can make out fine gradations: it's about sensitivity. Yes, some people can indeed make out the difference between 128 kbps MP3s and 256 kbps MP3s, and between these and CD quality. But then, as you've inferred, a higher bitrate means more space required on your hard disk or MP3 player—so it's a trade-off.

Talking about MP3, we move on to a discussion of digital audio formats: uncompressed, lossless compressed, and lossy compressed (MP3 falls in the last category because the 128 kbps means that the remaining bits are forever lost. Remember that if you rip a CD to MP3 and then throw the CD away, you *are* actually throwing something away!)

2.3 Uncompressed Formats

WAV (or WAVE)

This is a Microsoft and IBM audio file format standard for storing audio on PCs. It happens to be the native digital audio format in Windows. In WAV files, 8-bit or 16-bit samples are taken at rates of 11 kHz, 22 kHz or 44.1 kHz. The highest quality (16-bit at 44.1 kHz) uses 88 KB of storage per second of music. The WAV format is widely used as the medium for professional recording and editing. For creating audio CDs, WAV files are converted to the CDDA audio format. Both CDDA files and WAV files at their highest sampling rates take up a similar amount of storage space, and are not compressed (like MP3 is). But WAV is not the same as CDDA, as some people believe.

AIFF

Audio Interchange File Format is another format used for storing digital audio data. It supports a variety of bit resolutions, sample